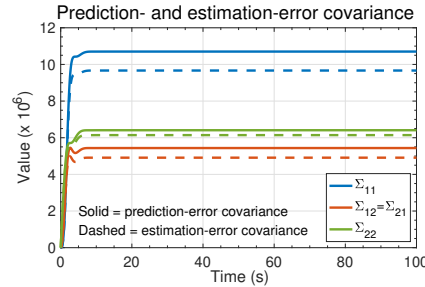




Convergence of the covariance matrices

- We sometimes observe an interesting phenomenon when we plot the prediction- and estimation-error covariance-matrix entries versus time.
- Often, they exhibit an initial transient and then converge to steady-state solutions.
- The figure to the right shows the entries of $\Sigma_{\tilde{x},k}^-$ and $\Sigma_{\tilde{x},k}^+$ for the spring-mass-damper system we have been using in our examples.
- After about 10 s (or 100 iterations), there is no longer any clear variation in these parameters.
- Since the primary purpose for computing $\Sigma_{\tilde{x},k}^-$ and $\Sigma_{\tilde{x},k}^+$ is to be able to compute L_k , this implies that if the covariances converge to a steady-state solution, L_k must also converge to a steady-state solution.



Idea

- Idea: Can we replace the time-varying L_k in the Kalman filter with a constant L_{ss} vector?
 - If yes, then we can omit all calculations of $\Sigma_{\tilde{x},k}^-$ and $\Sigma_{\tilde{x},k}^+$, which dramatically reduces the computational demand of the Kalman filter.
- Since the optimal solution uses L_k , a solution that uses L_{ss} instead will (by definition) be sub-optimal.
 - But, perhaps we are willing to accept a small reduction in estimate quality if the reduction in computation is significant.
- Conditions that guarantee the existence of a steady-state solution:
 - In addition to prior assumptions, A , B , C , and D are constant.
 - $\Sigma_{\tilde{w}} \geq 0$, $\Sigma_{\tilde{v}} > 0$, $\{C, A\}$ is “detectable” (or observable) and $\{A, \Sigma_{\tilde{w}}\}$ is “stabilizable” (or controllable).



Deriving the steady-state solution

- To derive a steady-state solution, we set:

$$\begin{aligned}\Sigma_{\tilde{x},k+1}^- &= \Sigma_{\tilde{x},k}^- = \Sigma_{\tilde{x},ss}^- \\ \Sigma_{\tilde{x},k+1}^+ &= \Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},ss}^+\end{aligned}$$

in the Kalman-filter steps, and solve.

- Consider filter loop as $k \rightarrow \infty$:

$$\begin{aligned}\Sigma_{\tilde{x},ss}^- &= A \Sigma_{\tilde{x},ss}^+ A^T + \Sigma_{\tilde{w}} \\ \Sigma_{\tilde{x},ss}^+ &= \Sigma_{\tilde{x},ss}^- - \underbrace{\Sigma_{\tilde{x},ss}^- C^T [C \Sigma_{\tilde{x},ss}^- C^T + \Sigma_{\tilde{v}}]^{-1} C \Sigma_{\tilde{x},ss}^-}_{L_{ss}}\end{aligned}$$

- Combine these two to get:

$$\Sigma_{\tilde{x},ss}^- = \Sigma_{\tilde{w}} + A \Sigma_{\tilde{x},ss}^- A^T - A \Sigma_{\tilde{x},ss}^- C^T [C \Sigma_{\tilde{x},ss}^- C^T + \Sigma_{\tilde{v}}]^{-1} C \Sigma_{\tilde{x},ss}^- A^T.$$



Solving this equation

- This equation is known as a “discrete algebraic Riccati equation” (DARE).
- For scalar systems (where the number of states $n = 1$), we can solve it using standard algebra, as we will do in an example in a few minutes.
- For systems having $n > 1$, we still have one matrix equation and one matrix unknown, so a solution is attainable.¹
- Octave implements a solution to the DARE in `dlqe.m`, which stands for “discrete linear quadratic estimator” (Kalman filters are optimal linear quadratic estimators).
 - Once we know $\Sigma_{\tilde{x},ss}^-$, we can compute $L_{ss} = \Sigma_{\tilde{x},ss}^- C^T [C \Sigma_{\tilde{x},ss}^- C^T + \Sigma_{\tilde{v}}]^{-1}$.
 - Error bounds can be computed via $\Sigma_{\tilde{x},ss}^+ = \Sigma_{\tilde{x},ss}^- - L_{ss} C \Sigma_{\tilde{x},ss}^-$.

¹A common algorithm used to solve a DARE is discussed in the paper: A. Laub, “A Schur method for solving algebraic Riccati equations.” *IEEE Transactions on Automatic Control*, 24(6), 1979, 913–921.



Scalar-system example

- To visualize, suppose: $x_{k+1} = x_k + w_k$; $z_k = x_k + v_k$.
- Let $\mathbb{E}[w_k] = \mathbb{E}[v_k] = 0$, $\Sigma_{\tilde{w}} = 1$, $\Sigma_{\tilde{v}} = 2$.
- Notice $A = 1$, $C = 1$, so:

$$\begin{aligned}\Sigma_{\tilde{x},ss}^- &= \Sigma_{\tilde{w}} + A \Sigma_{\tilde{x},ss}^- A - \frac{(A \Sigma_{\tilde{x},ss}^- C) (C \Sigma_{\tilde{x},ss}^- A)}{C \Sigma_{\tilde{x},ss}^- C + \Sigma_{\tilde{v}}} \\ &= 1 + \Sigma_{\tilde{x},ss}^- - \frac{(\Sigma_{\tilde{x},ss}^-)^2}{\Sigma_{\tilde{x},ss}^- + 2}\end{aligned}$$

$$\Sigma_{\tilde{x},ss}^- + 2 = (\Sigma_{\tilde{x},ss}^-)^2,$$

which leads to the solutions $\Sigma_{\tilde{x},ss}^- = -1$ or $\Sigma_{\tilde{x},ss}^- = 2$.

- We must choose the positive-definite solution, so $\Sigma_{\tilde{x},ss}^- = 2$ and $\Sigma_{\tilde{x},ss}^+ = 1$.

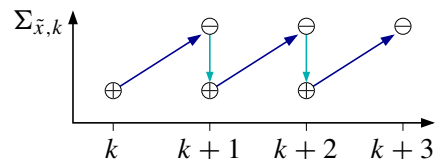


Verifying the example's solution

- If we iterate the covariance equations from some initial condition, we examine the evolution of $\Sigma_{\tilde{x},k}^-$ and $\Sigma_{\tilde{x},k}^+$ to see whether they agree with this solution:

$$\begin{array}{cccc}\Sigma_{\tilde{x},0}^- = \infty & \Sigma_{\tilde{x},1}^- = 3 & \Sigma_{\tilde{x},2}^- = 11/5 & \Sigma_{\tilde{x},3}^- = 43/21 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \Sigma_{\tilde{x},0}^+ = 2 & \Sigma_{\tilde{x},1}^+ = 6/5 & \Sigma_{\tilde{x},2}^+ = 22/21 & \Sigma_{\tilde{x},3}^+ = 86/85\end{array}$$

- Since uncertainty always increases during prediction and decreases during the measurement update, there will be two distinct steady-state covariances, as we have seen.
- The iteration does indeed converge toward $\Sigma_{\tilde{x},ss}^- = 2$ and $\Sigma_{\tilde{x},ss}^+ = 1$.





Implementing the steady-state filter in Octave

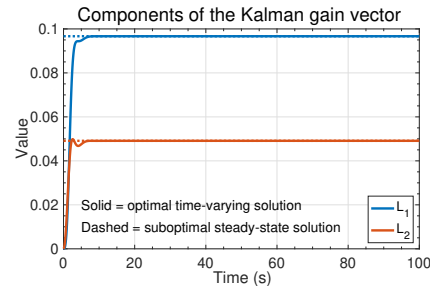
- It is simple to compute the steady-state covariances and Kalman gain in Octave:

```
[Lss, SigmaMinusss, SigmaPlusss] = dlqe(Ad, eye(nx), Cd, SigmaW, SigmaV);
```

- When we implement a steady-state Kalman filter, we can combine steps such that:

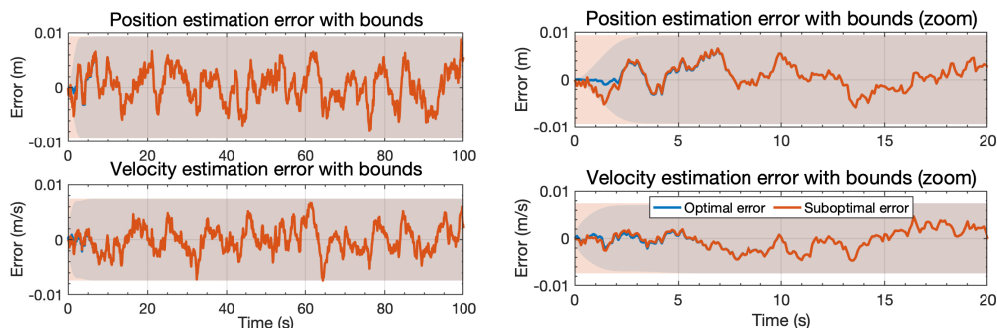
$$\hat{x}_k^+ = A\hat{x}_{k-1}^+ + Bu_{k-1} + L_{ss}(z_k - C(A\hat{x}_{k-1}^+ + Bu_{k-1}) - Du_k).$$

- We can use the precomputed steady-state $\Sigma_{\bar{x},ss}^+$ to output confidence bounds, so no online computation is required.
- The figure compares optimal L_k versus L_{ss} for the spring-mass-damper example.
- We see very little difference after about 10 s.



Results from steady-state filtering

- The figures compare estimation error and bounds between the optimal and suboptimal steady-state Kalman filters.
- The zoom plots on the right help us see that the differences are most evident in the first seconds of operation; after that, the filters perform essentially identically.



Summary

- Under some common conditions, the prediction-error and estimation-error covariances of a Kalman filter converge to steady-state solutions.
- In these cases, the Kalman gain also converges to a steady-state solution.
- If we replace the time-varying L_k in the Kalman filter with the steady-state L_{ss} , we no longer need to compute $\Sigma_{\bar{x},k}^-$ or $\Sigma_{\bar{x},k}^+$.
- This can save a lot of computation, but does result in a suboptimal solution.
- An example demonstrated that a steady-state solution differs most from the optimal solution only at the beginning of the filter's execution until the transient in L_k decays.
- If this level of error is acceptable, a steady-state Kalman filter can be a good option.